

A.3 Work, Energy and Power

Past Paper Questions

These are past paper questions selected from past papers (2016 - 2022) that relate to the topic A.3 Work, Energy and Power. Answers can be found in the separate accompanying document

Question 1

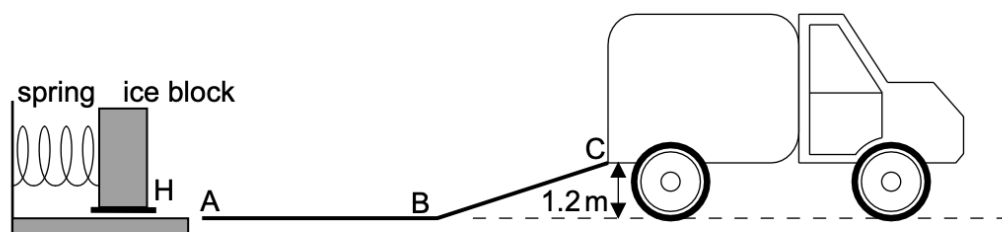
A stone is falling at a constant velocity vertically down a tube filled with oil. Which of the following statements about the energy changes of the stone during its motion are correct?

- I. The gain in kinetic energy is less than the loss in gravitational potential energy.
 - II. The sum of kinetic and gravitational potential energy of the stone is constant.
 - III. The work done by the force of gravity has the same magnitude as the work done by friction.
- A. I and II only
 - B. I and III only
 - C. II and III only
 - D. I, II and III

- **Learning Objective:** A.3.6
- **Level:** Standard Level
- **Past Paper:** May 2016, Paper 1
- **Question type:** Multiple Choice

Question 2

A company designs a spring system for loading ice blocks onto a truck. The ice block is placed in a holder H in front of the spring and an electric motor compresses the spring by pushing H to the left. When the spring is released the ice block is accelerated towards a ramp ABC. When the spring is fully decompressed, the ice block loses contact with the spring at A. The mass of the ice block is 55 kg.



Assume that the surface of the ramp is frictionless and that the masses of the spring and the holder are negligible compared to the mass of the ice block.

- (a) (i) The block arrives at C with a speed of 0.90 m s^{-1} . Show that the elastic energy stored in the spring is 670 J. [2]
- (ii) Calculate the speed of the block at A. [2]

- **Learning Objective:** A.3.5, A.3.6
- **Level:** Standard and Higher Level
- **Past Paper:** May 2016, Paper 2
- **Question type:** Calculate

Question 3

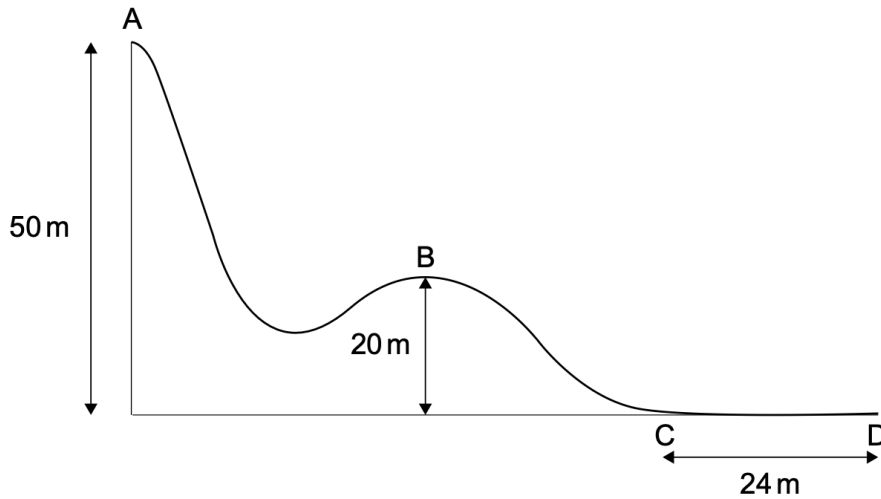
A student of weight 600 N climbs a vertical ladder 6.0 m tall in a time of 8.0 s. What is the power developed by the student against gravity?

- A. 22 W
- B. 45 W
- C. 220 W
- D. 450 W

- **Learning Objective:** A.3.7
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 4

The diagram below shows part of a downhill ski course which starts at point A, 50 m above level ground. Point B is 20 m above level ground.



A skier of mass 65 kg starts from rest at point A and during the ski course some of the gravitational potential energy transferred to kinetic energy.

From A to B, 24 % of the gravitational potential energy transferred to kinetic energy. Show that the velocity at B is 12 m s^{-1} .

[2]

- **Learning Objective:** A.3.5
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2017, Paper 2, TZ1
- **Question type:** Calculate

Question 5

A motor of input power 160 W raises a mass of 8.0 kg vertically at a constant speed of 0.50 m s^{-1} . What is the efficiency of the system?

- A. 0.63 %
- B. 25 %
- C. 50 %
- D. 100 %

- **Learning Objective:** A.3.8
- **Level:** Standard and Higher Level
- **Past Paper:** May 2018, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 6

An object falls from rest from a height h close to the surface of the Moon. The Moon has no atmosphere.

When the object has fallen to height $\frac{h}{4}$ above the surface, what is

$$\frac{\text{kinetic energy of the object at } \frac{h}{4}}{\text{gravitational potential energy of the object at } h} ?$$

- A. $\frac{3}{4}$
- B. $\frac{4}{3}$
- C. $\frac{9}{16}$
- D. $\frac{16}{9}$

- **Learning Objective:** A.3.5
- **Level:** Standard and Higher Level
- **Past Paper:** May 2018, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 7

The tension in a horizontal spring is directly proportional to the extension of the spring. The energy stored in the spring at extension x is E . What is the work done by the spring when its extension changes from x to $\frac{x}{4}$?

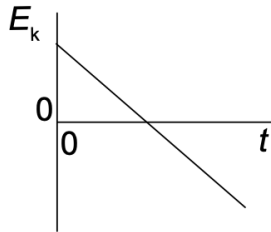
- A. $\frac{E}{16}$
- B. $\frac{E}{4}$
- C. $\frac{3E}{4}$
- D. $\frac{15E}{16}$

- **Learning Objective:** A.3.5
- **Level:** Standard and Higher Level
- **Past Paper:** Nov 2019, Paper 1
- **Question type:** Multiple Choice

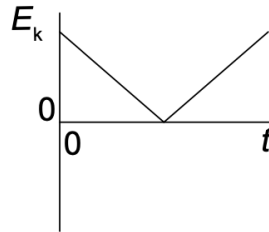
Question 8

A ball is thrown vertically upwards. Air resistance is negligible. What is the variation with time t of the kinetic energy E_k of the ball?

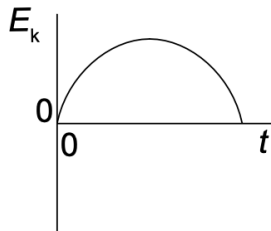
A.



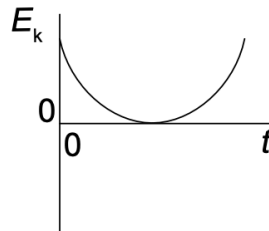
B.



C.



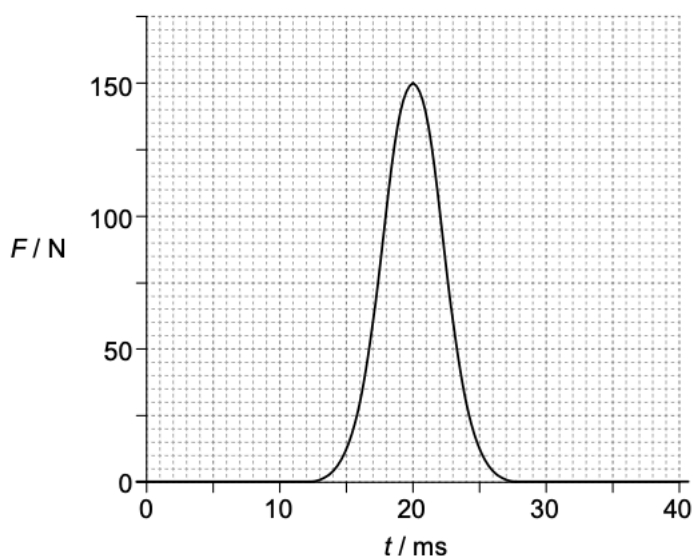
D.



- **Learning Objective:** A.3.5
- **Level:** Standard Level
- **Past Paper:** Nov 2019, Paper 1
- **Question type:** Multiple Choice

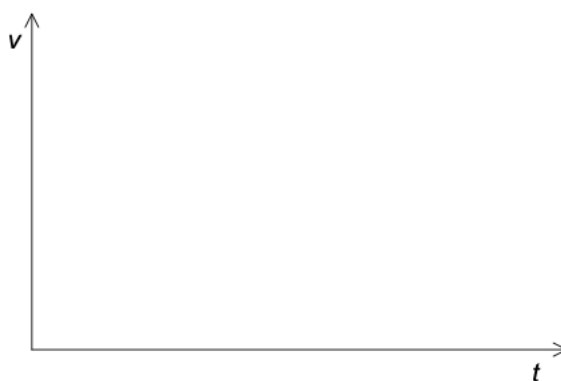
Question 9

The graph shows the variation with time t of the horizontal force F exerted on a tennis ball by a racket.



The tennis ball was stationary at the instant when it was hit. The mass of the tennis ball is $5.8 \times 10^{-2} \text{ kg}$. The area under the curve is 0.84 N s .

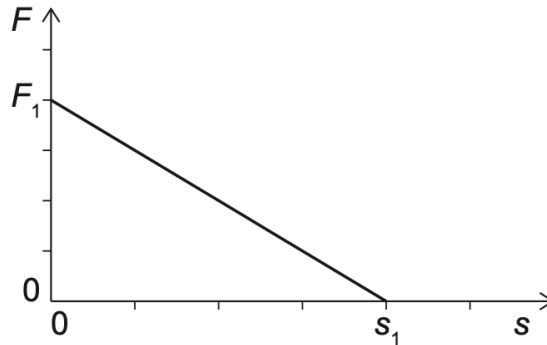
- (a) Calculate the speed of the ball as it leaves the racket. [2]
- (b) Show that the average force exerted on the ball by the racket is about 50 N. [2]
- (c) Determine, with reference to the work done by the average force, the horizontal distance travelled by the ball while it was in contact with the racket. [3]
- (d) Draw a graph to show the variation with t of the horizontal speed v of the ball while it was in contact with the racket. Numbers are **not** required on the axes. [2]



- **Learning Objective:** A.3.2
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2019, Paper 2
- **Question type:** Calculate, Graph

Question 10

A net force F acts on an object of mass m that is initially at rest. The object moves in a straight line. The variation of F with the distance s is shown.



What is the speed of the object at the distance s_1 ?

- A. $\sqrt{\frac{F_1 s_1}{2m}}$
- B. $\sqrt{\frac{F_1 s_1}{m}}$
- C. $\sqrt{\frac{2F_1 s_1}{m}}$
- D. $\sqrt{\frac{4F_1 s_1}{m}}$

- **Learning Objective:** A.3.4
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2021, Paper 1
- **Question type:** Multiple Choice

Question 11

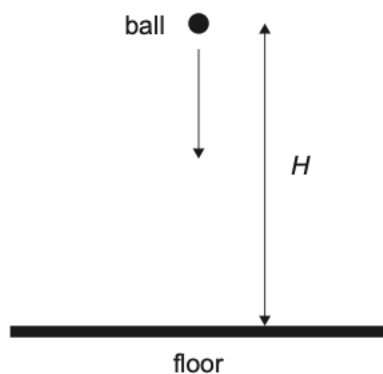
A cyclist rides up a hill of vertical height 100m in 500s at a constant speed. The combined mass of the cyclist and the bicycle is 80 kg. The power developed by the cyclist is 200W. What is the efficiency of the energy transfer in this system?

- A. 8%
- B. 20%
- C. 60%
- D. 80%

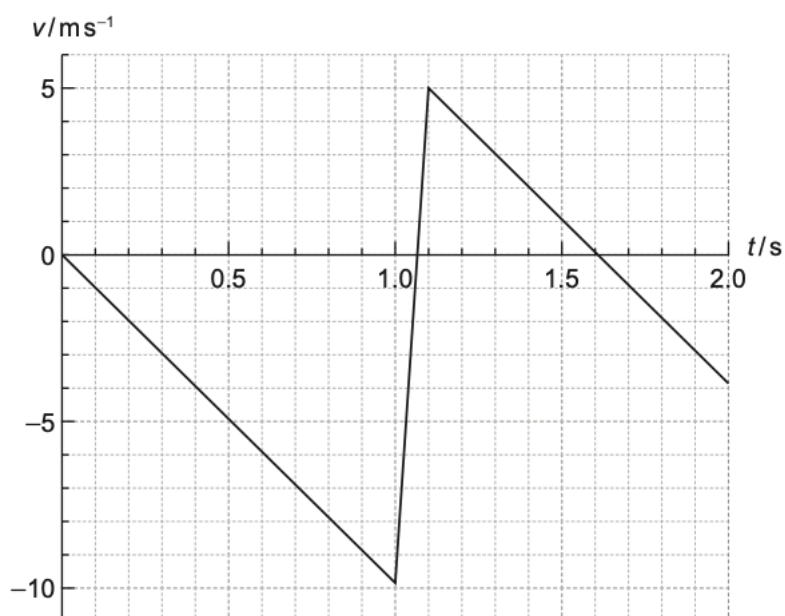
- **Learning Objective:** A.3.8
- **Level:** Higher Level
- **Past Paper:** Nov 2020, Paper 1
- **Question type:** Multiple Choice

Question 12

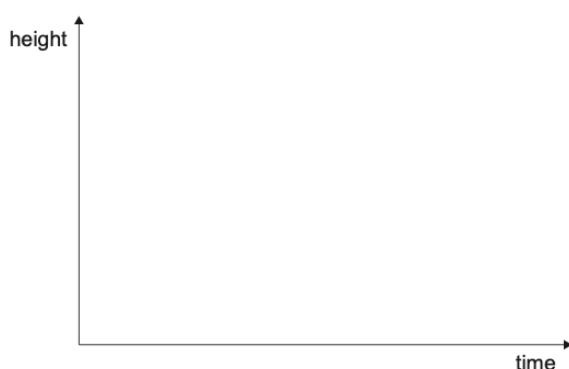
A ball of mass 0.250 kg is released from rest at time $t = 0$, from a height H above a horizontal floor.



The graph shows the variation with time t of the velocity v of the ball. Air resistance is negligible. Take $g = -9.80 \text{ ms}^{-2}$. The ball reaches the floor after 1.0 s .



- (a) Determine H . [1]
- (b) (i) Label the time and velocity graph, using the letter M, the point where the ball reaches the maximum rebound height. [1]
- (ii) State the acceleration of the ball at the maximum rebound height. [1]
- (iii) Draw, on the axes, a graph to show the variation with time of the height of the ball from the instant it rebounds from the floor until the instant it reaches the maximum rebound height. No numbers are required on the axes. [1]



- (c) Estimate the loss in the mechanical energy of the ball as a result of the collision with the floor. [1]
- (d) (i) Determine the average force exerted on the floor by the ball. [3]
- (ii) Suggest why the momentum of the ball was not conserved during the collision with the floor. [1]

- **Learning Objective:** A.3.5
- **Level:** Standard and Higher Level
- **Past Paper:** Nov 2021, Paper 2
- **Question type:** Draw, Calculate, Estimate, Explain

Question 13

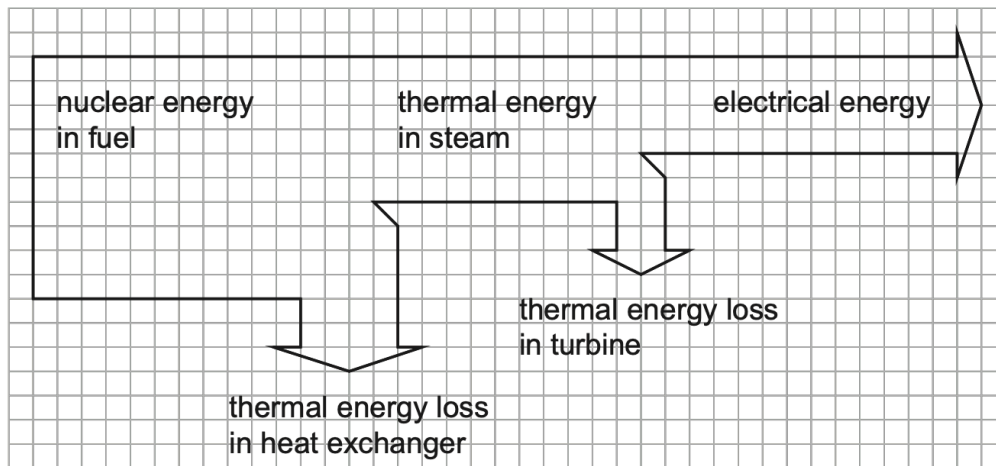
What is equivalent to $\frac{\text{specific energy of a fuel}}{\text{energy density of a fuel}}$?

- A. density of the fuel
- B. $\frac{1}{\text{density of the fuel}}$
- C. $\frac{\text{energy stored in the fuel}}{\text{density of the fuel}}$
- D. $\frac{\text{density of the fuel}}{\text{energy stored in the fuel}}$

- **Learning Objective:** A.3.9
- **Level:** Higher Level
- **Past Paper:** May 2018, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 14

The Sankey diagram shows the energy transfers in a nuclear power station.



Electrical power output of the power station is 1000 MW.

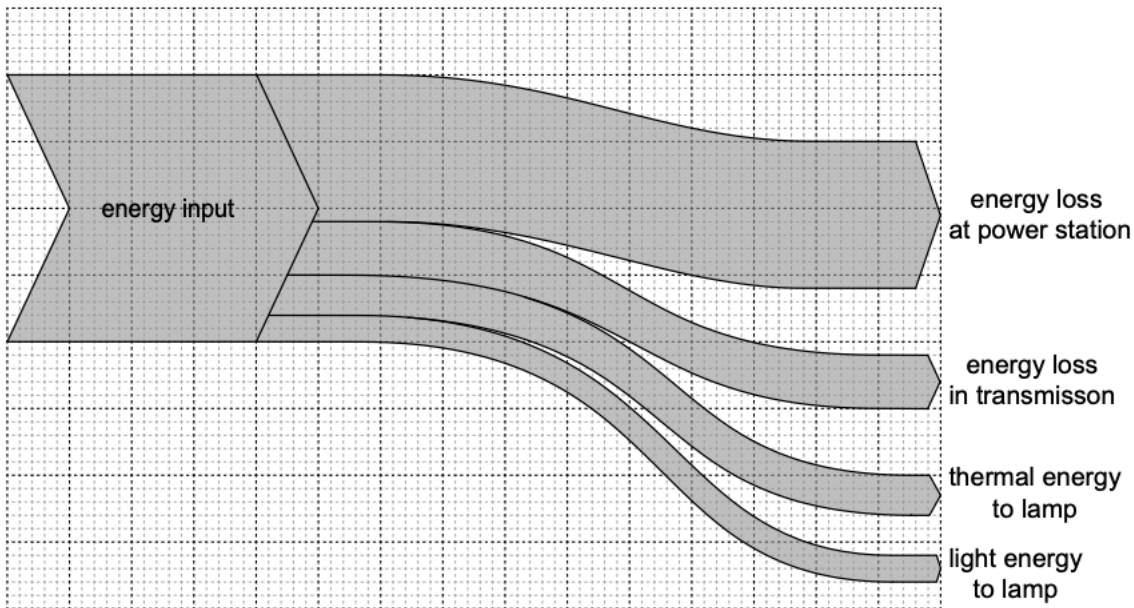
What is the thermal power loss in the heat exchanger?

- A. 500 MW
- B. 1000 MW
- C. 1500 MW
- D. 2500 MW

- **Learning Objective:** A.3.2
- **Level:** Standard Level
- **Past Paper:** Nov 2021, Paper 1
- **Question type:** Multiple Choice

Question 15

The Sankey diagram shows the energy input from fuel that is eventually converted to useful domestic energy in the form of light in a filament lamp.



What is true for this Sankey diagram?

- A. The overall efficiency of the process is 10 %.
- B. Generation and transmission losses account for 55 % of the energy input.
- C. Useful energy accounts for half of the transmission losses.
- D. The energy loss in the power station equals the energy that leaves it.

- **Learning Objective:** A.3.2
- **Level:** Higher Level
- **Past Paper:** May 2018, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 16

An engine is exerting a horizontal force F on an object that is moving along a horizontal surface at a constant velocity v . The mass of the object is m and the coefficient of dynamic friction between the object and the surface is μ .

What is the power of the engine?

- A. $\frac{Fv}{\mu}$
- B. μFv
- C. $\frac{mgv}{\mu}$
- D. μmgv

- **Learning Objective:** A.3.7
- **Level:** Higher Level
- **Past Paper:** Nov 2022, Paper 1
- **Question type:** Multiple Choice

Question 17

What is the definition of the SI unit for a force?

- A. The force required to accelerate, in the direction of the force, a mass of 1 kg at 1 m s^{-2}
- B. The force required to accelerate, in the direction of the force, a mass at 1 m s^{-2}
- C. The weight of a mass of 0.1 kg
- D. The change in momentum per second

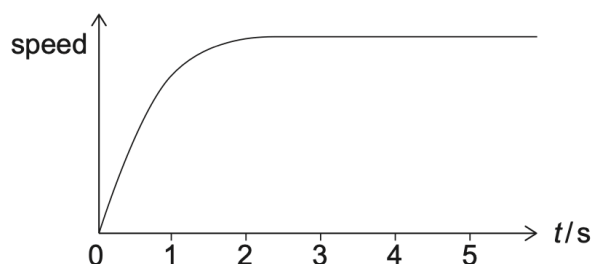
- **Learning Objective:** A.2 Math Skill 6 (I've sneaked this one in here!)
- **Level:** Higher Level
- **Past Paper:** Nov 2022, Paper 1
- **Question type:** Multiple Choice

Question 18

A raindrop falls vertically from rest.

- (a) State the initial acceleration of the raindrop. [1]

The graph shows how the speed of the raindrop varies with time t .



- (b) Explain, by reference to the vertical forces, how the raindrop reaches a constant speed. [3]
- (c) During the first 3.0 s of motion, the raindrop falls a distance of 21 m and reaches a speed of 9.0 m s^{-1} . The mass of the raindrop is 34 mg. The temperature of the raindrop does not change.
- (i) Determine the energy transferred to the air during the first 3.0 s of motion. State your answer to an appropriate number of significant figures. [3]
- (ii) Describe the energy change that takes place for $t > 3.0 \text{ s}$. [1]

Question 19

A solar farm is made up of photovoltaic cells of area $25\,000\text{ m}^2$. The average solar intensity falling on the farm is 240 W m^{-2} and the average power output of the farm is 1.6 MW . Calculate the efficiency of the photovoltaic cells.

[2]

Question 20

The water in a particular pumped storage hydroelectric system falls a vertical distance of 270 m to the turbines. Calculate the speed at which water arrives at the turbines. Assume that there is no energy loss in the system.

[2]